CRYPTOCURRENCY PRICE TRACKER

POST PROJECT REPORT

Automated Real-Time Crypto Data Extraction and Analysis

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Project Type: Individual Project

EXECUTIVE SUMMARY :

The Cryptocurrency Price Tracker is a Python-based automation project developed to collect real-time cryptocurrency market data from CoinMarketCap. The system uses Selenium for browser automation to extract the top 10 cryptocurrencies along with their price, 24-hour percentage change, and market capitalization. The extracted data is processed using Pandas and stored automatically in an Excel file for further analysis. This project demonstrates practical implementation of web scraping, data processing, and file handling using Python in a real-time financial monitoring scenario.

PROJECT OVERVIEW :

Objective:

To design and implement an automated cryptocurrency monitoring system that collects live market data from an online platform, records time-based entries for reference, and generates organized output files to support analysis and informed decision-making.

Scope:

• Focused on CoinMarketCap platform
• Extracts real-time cryptocurrency details
• Exports data in structured file format
• Maintains timestamp-based records
• Supports basic filtering options

Key Features:

• Real-time cryptocurrency data scraping
• Handles dynamic webpage content using Selenium
• Extracts top 10 coin details
•Structured file export for analysis
• Background (headless) execution support

TECHNICAL IMPLEMENTATION:

Architecture:

The system follows a sequential workflow:

1. Launch Chrome browser using Selenium WebDriver.
2. Navigate to the CoinMarketCap homepage.
3. Identify the cryptocurrency table elements.
4. Extract the top 10 coin details from the webpage.
5. Add a timestamp to record when the data was collected.
6. Store the data in a structured format using Pandas.
7. Export the final data into an Excel (.xlsx) file.

Technology Stack:

- Python 3.x – Used to write the automation script.
- Selenium – Handles browser automation and dynamic content scraping.
- Pandas – Used to organize and export data into Excel.
- WebDriver Manager – Automatically manages ChromeDriver installation.
- Datetime Module – Generates date and time information.

Performance Metrics:

- **Processing Capacity:**Extracts top 10 cryptocurrencies per execution.
- **Execution Time:**Approximately 5–10 seconds depending on internet speed.
- **Reliability:** High accuracy when webpage structure remains unchanged
- **Output Format:**Structured Excel file with properly labeled columns.

SOURCE CODE IMPLEMENTATION (Screenshot):

Run Terminal Help crypto_tracker.py

Untitled-1 crypto_tracker.py

crypto_tracker.py > ...

```python
# Cryptocurrency Price Tracker
# Author: Your Name
# Description: Scrapes Top 10 Cryptocurrencies from CoinMarketCap
# and stores data into Excel file.

import time
import pandas as pd
from datetime import datetime
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from webdriver_manager.chrome import ChromeDriverManager

# Setup Chrome Driver
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))

# Open Website
driver.get("https://coinmarketcap.com/")
time.sleep(5)  # Wait for page to load

# Extract Data
rows = driver.find_elements(By.XPATH, "//table/tbody/tr")

crypto_data = []

for i in range(10):  # Top 10 coins
    cols = rows[i].find_elements(By.TAG_NAME, "td")

    name = cols[2].text
    price = cols[3].text
    change_24h = cols[4].text
    market_cap = cols[7].text

    crypto_data.append({
        "Timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "Name": name,
        "Price": price,
```

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```python

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    cols = rows[i].find_elements(By.TAG_NAME, "td")

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    crypto_data.append({
        "Timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "Name": name,
        "Price": price,
        "24h Change": change_24h,
        "Market Cap": market_cap
    })

# Close Browser
driver.quit()

# Convert to DataFrame
df = pd.DataFrame(crypto_data)

# Save to Excel File
file_name = "crypto_prices.xlsx"
df.to_excel(file_name, index=False)

print("✅ Data successfully saved to", file_name)

```

Overview of Extracted Data :

	A	B	C	D	E	F
1	Timestamp	Name	Price	24h Change	Market Cap	
2	2026-02-22 16:57:10	CoinMarketCap 20 Index [	$139.78	0.07%	$13,381,088	
3	2026-02-22 16:57:10	BitcoinBTCBuy	$68,222.71	0.30%	$1,363,494,156,609	
4	2026-02-22 16:57:10	EthereumETHBuy	$1,981.96	0.34%	$239,028,833,784	
5	2026-02-22 16:57:10	TetherUSDTBuy	$0.9998	0.01%	$183,619,045,343	
6	2026-02-22 16:57:10	XRPXRPBuy	$1.42	0.02%	$86,852,149,219	
7	2026-02-22 16:57:10	BNBBNBBuy	$624.11	0.03%	$85,104,162,749	
8	2026-02-22 16:57:11	USDCUSDCBuy	$1.00	0.01%	$74,455,397,608	
9	2026-02-22 16:57:11	SolanaSOLBuy	$85.39	0.27%	$48,542,997,969	
10	2026-02-22 16:57:11	TRONTRXBuy	$0.2889	0.07%	$27,370,438,691	
11	2026-02-22 16:57:11	DogecoinDOGEBuy	$0.09753	0.14%	$16,468,395,234	
12						
13						

KEY ACHIEVEMENTS:

Technical Achivements:

- Successful automation of dynamic webpage using Selenium.
- Real-time data extraction without manual effort.
- Automatic Excel file generation.
- Organized and readable output format.

Process Automation Benefits:

- Saves time compared to manual tracking.
- Reduces chances of human error.
- Provides structured data for analysis.
- Can be extended for continuous monitoring.

LIMITATIONS AND CONSTRAINTS:

Current Limitations:

- Dependent on internet connectivity.
- Website structure changes may require code modification.
- Limited to top 10 cryptocurrencies.
- Requires Chrome browser and required libraries installed.

Resource Constraints:

- Execution speed depends on system performance.
- Dynamic content loading may slightly delay scraping.

PROJECT IMPACT AND VALUE:

Immediate Benefits:

- Automated collection of cryptocurrency data.
- Structured and clean Excel output.
- Improved efficiency in data monitoring.

Long-Term Potential:

- Can be expanded into a full crypto monitoring system.
- Can integrate with visualization tools like dashboards.
- Can be scheduled for daily or hourly tracking.
- Useful for investment analysis and trend monitoring.

CONCLUSION:

The Cryptocurrency Price Tracker project successfully demonstrates the practical use of Selenium and Pandas for real-time data extraction and storage. The system automates cryptocurrency market data collection and stores it in Excel format for further use. This project highlights the effectiveness of automation in financial data monitoring and provides a strong foundation for future enhancements.